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Multimodal assessment of political corruption worsening national poverty: Evidence of mediating and moderating effects from global panel data

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Abstract

Political corruption is one of the structural factors aggravating national poverty, on which there is a consensus. However, it remains controversial what factors constrain the causal mechanisms by which political corruption worsens national poverty. This paper analyzed this using different models with the global panel data of 117 counties from 1992 to 2017, including the instrumental variable model. The authors found that: (1) There was a significant positive correlation between political corruption and national poverty, (2) Political corruption exacerbating national poverty was constrained by primary education and the Gini index, which played a mediating role, (3) The level of foreign direct investment and public services moderated the relationship between political corruption and national poverty. Finally, the authors discussed the potential implications beyond national poverty for theory, method, and policy.

Keywords

Instrumental variable model, mediating effects, moderating effect, national poverty, political corruption

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Introduction

Poverty is a global issue. According to data from the United Nations, over 780 million people live in extreme poverty. Significant populations who live in Africa, Latin America, and South Asia are thus adversely affected. National poverty has become a significant obstacle to individual and national development. For individual development, the poor lack the resources needed for survival and fall into a situation of malnutrition, inadequate medical care, and unmet educational needs. The poor cannot receive higher pay in the labor market due to their skills' restrictions. Other advantaged social groups (or the upper class) can benefit more quickly from economic development than the poor. Regarding national development, African countries with high poverty ratios have less access to resources globally, and the gap with rich countries widens. Hence, national poverty alleviation is a crucial global sustainable development goal.

Similar to national poverty, political corruption is also a common social phenomenon and a structural governance issue that has drawn more and more attention from scholars. The extant literature has found that political corruption not only hinders economic growth and drags down people's income, but also cuts expenditure on public services and social security, erodes social support for the poor, leaving them deeply mired in national poverty (Chetwynd et al., 2003). In addition, political corruption shakes the general political trust between individuals and governments (Rothstein & Uslaner, 2005), exacerbating the social exclusion of the poor. However, there is still a lack of consensus on the causal mechanism of political corruption and national poverty, and there is still room for discussion in empirical research from a global perspective. This

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study explores the issue based on the panel data of 117 countries from 1992 to 2017.

The paper is developed as follows: the first part is an introduction, leading to the research questions of this paper; part two is a literature review, explaining the research nexus between political corruption and national poverty in existing research and developing the research hypotheses; part three is the introduction of data and methods; part four is the results of the empirical analysis, and finally, the discussion and conclusions.

Literature review

This part mainly expounds on the complex relationship between political corruption and national poverty, as well as the economic model, governance model, education model, and polarization model that political corruption worsens national poverty. Accordingly, the authors develop the research hypotheses of this paper.

Structural elements are the leading causes of national poverty

National poverty has been one of the critical issues of social science for a long time. Poverty research has shifted from a single perspective to a multidimensional perspective in academia. In particular, the definition of poverty under a single perspective is mainly from economics; as living beneath the poverty line is a criterion to be considered, it considers people whose household income cannot meet the basic living needs (Rowntree, 1902). Then Amartya Sen's capacity poverty theory is representative of the multidimensional perspective, which holds that poverty is caused by depriving individuals of substantive freedom and essential abilities (Sen, 1982). The poor cannot interact with the external environment to achieve sustainable self-development.

In the early 20th century, when scholars evaluated poverty from the economic status of individuals or families, they considered it a unique phenomenon, not a social problem (Laderchi et al., 2003). However, poverty has increasingly evolved into a structural social problem with the development of industrialization and postmodernism at the end of the 20th century. More and more scholars are looking for structural factors that lead to poverty. For example, the abuse of power for personal benefits in a bureaucratic system squeezes the living space of the poor who lack political attention and political discourse (Chattopadhyay, 2013). Political corruption is a crucial part of developing poverty reduction policies in countries.

Political corruption is an essential structural element

There is abundant research concerning political corruption in academic circles. In terms of their behaviors, political corruption refers to the phenomenon that political activists, politicians, or officials involved in the decision-making process of judges make use of the information and convenience they hold as public officials to seek private interests in exchange for actual rewards that is, the non-public use of public power (Svensson, 2005). From the perspective of cost and benefit, it is considered that political corruption is the exchange of power and money, and a few people choose the lower bribery cost ratio to obtain higher income or excess profits, which is the behavior of rent setting and rent seeking (Aidt, 2003; Klitgaard, 1998). As a behavior anomie, the pursuit of social resources for individuals must follow certain social norms. Political corruption is regarded as a negative deviant behavior because it violates reasonable and legal social norms and impedes public social life or social progress (Roberts, 2015).

Accordingly, political corruption mainly refers to political activists or officials who take advantage of their information and convenience in public offices to break the rules for private interests or exchange public power for illegal remuneration in decision-making. Political corruption is a common social phenomenon and a structural governance issue (Alatas, 2015).

The complex relationship between political corruption and national poverty

Scholars have pointed out that political corruption is often accompanied by national poverty, and the causal relationship between the two is intricate (Lambsdorff, 1999). Some took dynamic panel data of 97 developing countries from 1997 to 2006 to conduct Gaussian Mixture Model (GMM) estimation. They found that political corruption and national poverty were mutually reinforcing and proposed comprehensive strategies to combat political corruption and reduce national poverty (Negin et al., 2010).

Beyond the empirical quantitative data, there is a persistent debate about how political corruption affects national poverty research, and whether the spillover effect of political corruption can help reduce national poverty is controversial.

Spillover effects of political corruption: Whether political corruption worsens national poverty

Scholars represented by Huntington (2006) regarded political corruption as the "grease the wheel," believing that traditional laws or bureaucratic systems impede economic development in developing countries. Bribery allows companies to operate effectively and reduces efficiency losses caused by government meddling (Ahlin & Bose, 2007). At the same time, political corruption also enables decision-makers to obtain more information from informal channels to avoid the government's fault in decision-making to a certain extent (Bayley, 1966). Therefore, the scholars

supporting this opinion assumed that moderate corruption was a mercy. Scholars argued that though violence and bribery were illegal in the power system, the latter were more peaceful and could regulate social conflicts at a lower cost (Leff, 1964; Leys, 1965).

However, scholars took the opposite view and regarded it as "sand on wheels" (Méon & Sekkat, 2005). They challenged the greasing wheel thinking and found that far from greasing the wheels of commerce, more political corruption means even more red tape (Wei & Kaufmann, 1999). Researchers believed that political corruption increased transaction costs of government and enterprises (Mauro, 1995) and brought high development costs to developing countries (Shleifer & Vishny, 1993). It reduced the quality of government and governance efficiency (Gilley, 2006), which led the decline of the legitimacy of a country's political system both domestically and worldwide (Rothstein & Teorell, 2008). In addition, it undercuts efforts to protect civil political rights (Alston & Robinson, 2005), especially making the initially vulnerable poor people fall into "aphasia" regarding their survival and development.

The occurrence pattern and research hypothesis of political corruption worsening national poverty

Currently, it has become the consensus of scholars that political corruption worsens national poverty and exacerbates national poverty in almost all cases (Serritzlew et al., 2014). Accordingly, the first research hypothesis of this paper is proposed:

> H1: *Ceteris paribus*, from a global perspective, an increase in political corruption in a country can significantly raise its national poverty level.

Different paths for its formation are generally confirmed, the specific patterns of political corruption worsening national poverty will be analyzed below, and the corresponding moderating and mediating effects' hypotheses will be proposed accordingly.

Firstly, political corruption worsens the poor's educational model. Scholars advocate using education to fight national poverty (Tilak, 2002). A lower level of education means a lower quality workforce, unable to earn higher wages, and scholars have found that the popularization of primary education has significantly reduced poverty (Song, 2012). In contrast, political corruption worsens the development of primary education, indirectly exacerbating national poverty. This means that the labor force is less skilled and can only do lower-wage jobs, such as manual labor (Xiao & Bian, 2018). However, political corruption makes educational resources more inclined to the privileged and elite class, and the quantity and quality of primary education worsen the development of primary education, indirectly exacerbating national poverty. The needs of the poor cannot be fully satisfied. Therefore, we propose the following hypothesis:

> H2a: *Ceteris paribus*, political corruption worsens the development of primary education, indirectly exacerbating national poverty.

Secondly, political corruption worsens the polarizing model of national poverty. Political corruption aggravates social inequality, leading to the accumulation of social wealth for the rich, while the poor are increasingly disadvantaged regarding income distribution, leading to a decrease in income and an increase in inequality (Gupta et al., 2002). Thus, political corruption can affect national poverty by affecting social inequality. Accordingly, we propose the following hypothesis:

> H2b: *Ceteris paribus*, political corruption increases social inequality, indirectly exacerbating national poverty.

Thirdly, political corruption worsens national poverty through the economic model. In the extant studies, political corruption exerts an impact on national poverty through the following processes: on the one hand, political corruption increases the cost of foreign direct and domestic investment, reduces the scale of domestic investment, and narrows the scope of investment. Empirical studies have shown that the higher the level of political corruption, the lower the level of foreign investment (Campos et al., 1999). Domestic investors find it challenging to carry out high-quality business activities because of the high hidden costs and poor primary investment context (Beekman et al., 2014). Furthermore, it leads to slow economic development so that the income of the poor falls. Accordingly, we propose the following hypothesis:

> H3a: *Ceteris paribus*, political corruption is more likely to worsen national poverty in countries with low levels of foreign investment.

Finally, political corruption worsens national poverty through governance models. Political corruption and low public services always go hand in hand in the development of a country. Low public services means that the basic demands of the poor cannot be answered in time, which exacerbates their poverty level. Scholars have found that poor national governance capacity reduced the supply of public goods which the poor need urgently (Aaberge et al., 2010; Anokhin & Schulze, 2009), which also reduces production motivation and further increases livelihood uncertainty and vulnerability of the poor (Berkovich, 2016; Jindra & Vaz, 2019), thus exacerbating national poverty. However, political corruption is an essential factor that deteriorates public services. Therefore, we propose the following hypothesis:

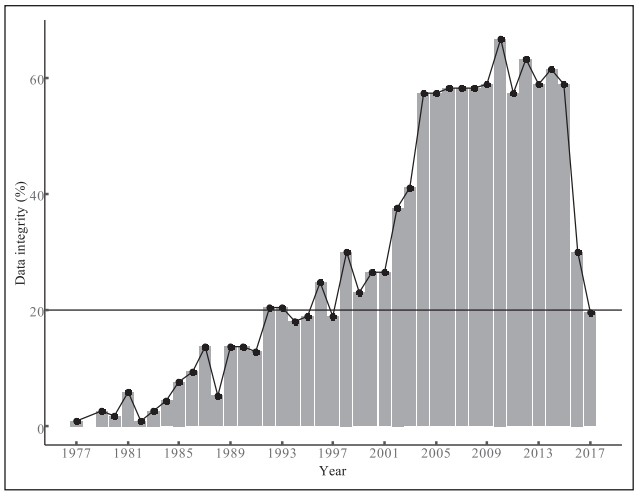

Figure 1. Data completeness of poverty incidence in period time.

H3b: *Ceteris paribus*, political corruption is more likely to worsen national poverty in countries with levels of public services.

Data and models

This part introduces this paper's data selection and model construction, including how the samples were screened according to secondary data, how independent and dependent variables were measured, how control and other important variables were selected, and how models were constructed.

Data resources

This paper adopted the Quality of Government data (QoG) from the Institute of Quality of Government of the University of Gothenburg in Sweden. QoG covers cross-section and time-series data in politics, economy, society, and culture in more than 200 countries and regions from 1947 to 2020.

National poverty was the explained variable. Considering the feasibility of the data, the authors deleted the countries with only two or less than two data sets from 1977 to 2020, yielding 128 samples of countries (as shown in Supplemental Appendix Table A1). In terms of a time dimension, Figure 1 shows the data integrity of poverty incidence from 1977 to 2020. The paper selected the panel data of 117 countries in the QoG database from 1992 to 2017 to study the influencing mechanism of political corruption on national poverty incidence, considering the data availability.

Variable selection and data description

Political corruption and national poverty were independent variables and dependent variables, respectively. The method of national poverty measurement was based on two dimensions: absolute poverty and relative poverty. The former argues that whether the family income meets the basic needs is a critical threshold in dividing the rich and the poor. Currently, the World Bank's standard of absolute poverty is accepted by academics, which defines people living on less than $1.90 per person per day as extremely poor. In low-and middle-income countries, those living on less than $3.20 per person per day are poor. Furthermore, in middle- and high-income countries, the poor live on $5.50 per person per day (Jolliffe & Prydz, 2016; World Bank, 2018).

Regarding measuring relative poverty, the commonly selected indicators for measuring relative poverty are the human poverty index, the human development index, and the multidimensional poverty index. However, the selection of relative poverty indicators is highly subjective, making it difficult to reflect absolute poverty globally. Therefore, this paper chose the $3.20 per person per day poverty standard of the World Bank to reflect the changes in the global poverty population, and $1.90 per person per day was used for the substitution variable test.

There are commonly used measures of political corruption, such as the corruption perceptions' index (CPI) of Transparency International, the control of corruption indicator (CCI) of the World Bank, and the political corruption index (PCI) of the V-Dem Institute. The CPI and CCI focus on measuring the respondents' perception of corruption rather than corruption itself, and the variable is an unbalanced dataset and cannot fully and objectively reflect the level of corruption in a country (Mungiu-Pippidi & Dadašov, 2016; Transparency International, 2018). The PCI incorporates six types of corruption measures, covering different dimensions of the polity, including executive, legislative, and judicial corruption. This paper chose PCI as an operational indicator of corruption (Coppedge et al., 2021).

The paper also chose the following control variables based on the extant literature. Firstly, economic development is the cornerstone for the income growth of the poor, and sufficient aggregate growth can benefit the poor in all dimensions (Agrawal, 2007). Countries with a higher economic growth rate can reduce national poverty more effectively (Montalvo & Ravallion, 2010). Therefore, we used the annual growth rate of per capita GDP to measure economic development.

Secondly, financial inclusion is a critical factor for inclusive growth. Countries with higher financial inclusion are more likely to have low-income groups covered by financial services, thus reducing the incidence of national poverty and income inequality (Park & Mercado, 2015). Studies on tax burden have found that tax relief is conducive to national poverty alleviation instead of tax increases, and income tax credits reduce the burden on low-income families (Simon et al., 2018).

Thirdly, population and social factors are significant causes of national poverty and famine (Fernihough & O'Grada, 2018). Researchers have found that population pressure and density negatively impact rural poverty (Wang et al., 2018). Furthermore, the extant literature indicates that unemployment ramps up poverty and inequality and

negatively influences laid-off individuals and their families and communities (Akwara et al., 2013). Education expenditure and electricity services could reduce national poverty effectively and promote long-term economic development (Jung & Thorbecke, 2003; Ogun, 2010).

Finally, the World Bank has found that the quality of agricultural land indirectly impacts income growth through a study on rural land usage in 83 developing countries. Therefore, the distribution of pastoral resources represented by the proportion of agricultural land closely correlates with national poverty (Barbier & Hochard, 2014). In addition, scholars have illustrated that the development of the service sector can contribute to poverty reduction (Benfica & Henderson, 2021), and we chose the employment in services to measure service economy. In addition to the primary and control variables mentioned above, this paper introduced other variables for the endogenous and robustness tests. Net inflows of foreign direct investment, public services, the Gini index, and primary school enrollment were applied to conduct mechanism analysis to increase the model's credibility. This paper discussed the regression estimation's consistency to further test the robustness of the estimation by using the variable substitution and instrumental variables' model, respectively. Additionally, the authors took "a person living on $3.20 per day" as the substitute variable for the dependent variable to compare and analyze the estimates of the original model. Patents' grants and the lag of political corruption were instrumental variables in the analysis model. The statistical description of all variables is shown in Table 1.

The authors also provide a scatter plot (Figure 2) of political corruption and national poverty. Figure 2 preliminarily shows the relationship between political corruption and national poverty, and the impact of political corruption on national poverty appears in a different reversed U trend during a different period. It shows that with the improvement of political corruption in a country, national poverty will be aggravated, laying a foundation for further empirical analysis in the following sections.

Models

Multiple regression. The multivariate variables' model was established based on the classical linear model to explain the mentioned hypotheses, as shown in Equation 1:

$$Poverty_gap = \beta_0 + \beta_1 vdem_corr + \beta_2 Controls + \varepsilon \quad (1)$$

$$\varepsilon \sim N(0, \delta^2)$$

where, $\beta_0, \beta_1, \beta_2$ are denoted regression coefficients, and ε represents the error term, obeying a normal distribution of the mean value of 0 and standard deviation of δ. The maximum likelihood estimations (MLE) were applied in the statistical analysis of Equation 1 to return the estimated value of the regression coefficient. Multiple regression analysis ignored the time (year) trend. Here, the paper introduced a feasible generalized least squares (FGLS) estimation method, which explained the time trend of the data and solved the heteroscedasticity in the multiple regression model. Therefore, the FGLS method in the panel data was significantly better than the mixed regression in the multiple regression model. The *F* test confirmed the finding.

Fixed effects' model. According to the effects of countries, the following format used a simple fixed effects (FE) model:

$$Poverty_gap_{it} = \beta vdem_corr_{it} + \delta Controls_{it} + \alpha_i + p_t + \varepsilon_{it} \quad (2)$$

where i is country, $i = 1. . ., N$; t is a year, $t = 1992. . . 2017$; $Poverty_gap_{it}$ is the explanatory variable (poverty gap of $3.20 per day in 2011 PPP (%)); $vdem_corr_{it}$ is a variable varying with time, which represents the degree of political corruption control; β is the corresponding coefficient; $Controls_{it}$ is the matrix of the control variable, and δ is the corresponding coefficient; α_i is the fixed country effect, which is related to an explanatory variable. p_t is the fixed time effect (year); ε_{it} represents the vectors of the error term, assuming that they are exogenous:

$$\sum\left(\varepsilon_{it} | \alpha_i, x_{it}\right) = 0, i = 1, 2, \ldots\ldots N, t = 1992, \ldots, 2017 \quad (3)$$

Many methods, such as the traditional maximum likelihood estimator, estimated the coefficients in the FE model. In addition, scholars proposed a transformed likelihood method test with a maximum likelihood estimator and minimum distance estimator to surmount the accidental parameters that violate MLE's consistency and asymptotic efficiency (Hsiao et al., 2002). Since the data set used in this paper was characterized by a relatively large N yet short T, the transformed likelihood method was chosen. What is more, the random effect (RE) model was consistent with Equation 2, but α_i, the individual trend in the further hypothesis was not correlated with x_{it} the explanatory variable. The difference between different individuals $i = 1, 2, \ldots\ldots N$ was random and obeyed normal distribution.

$$corr\left(\alpha_i, x_{it}\right) = 0 \quad (4)$$

The RE and FE model ae the most selected panel data models. The applicability of the two is usually considered as being hinged on the Hausman (1978) test to verify whether the unique error term (ε_{it}) is correlated with the regression variables, and the null hypothesis is unrelated.

Table 1. Data statistical description.

Variable type	Variable define	Obs.	Mean	Std. Dev.	Min	Max
Explained variable	Poverty gap at USD 3.20 a day (2011 PPP) (%)	1,337	9.12	13.60	0	77.10
Explanatory variables	Political corruption index	4,424	0.53	0.30	0.01	0.98
Control variables	Tax burden	3,694	72.37	15.62	0	100
	Investment freedom	3,701	52.53	21.00	0	95
	Unemployment, total (% of total labor force) (national est.)	2,398	8.48	6.03	0.14	38.80
	Government expenditure on education, total (% of government expenditure)	2,342	14.84	5.07	0.85	47.28
	Access to electricity (% of population)	4,239	77.25	31.31	0.01	100
	GDP growth (annual %)	4,748	3.75	6.41	−62.08	149.97
	Population density (people per sq. km of land area)	4,896	257.50	1,321.87	1.44	19,196
	Financial freedom	3,693	49.69	19.86	0	90
	Arable land (% of agricultural land)	4,654	38.73	26.46	0.06	100
	Population, total	4,945	33.8×10^6	129×10^6	9,112	$1{,}390 \times 10^6$
	Employment in services (% of total employment) (modeled ILO)	4,567	49.00	18.68	5.30	87.85
Lagged variable	Poverty gap at USD 3.20 a day (2011 PPP) (%) lag one period	1,313	9.22	13.69	0.00	77.10
	Poverty gap at USD 3.20 a day (2011 PPP) (%) lag two periods	1,276	9.28	13.73	0.00	77.10
Moderating variables	Foreign direct investment, net inflows (% of GDP)	4,584	6.67	40.88	−58.32	1,282.63
	Public services	2,263	5.69	2.45	0.70	10
Mediating variables	GINI index (World Bank estimate)	1,334	39.13	9.35	23.70	65.80
	School enrollment, primary (% gross)	3,969	100.95	16.67	20.88	165.65
Substitution variables	Poverty gap at USD 1.90 a day (2011 PPP) (%)	1,337	4.23	8.05	0	63.60
Instrument variables	Political corruption index lag one period	4,251	0.53	0.30	0.01	0.98
	Patent grants	3,640	875.16	7,076.93	0	150,952

Empirical analysis

Baseline regression results

Based on the data sets of 117 countries from 1992 to 2017, this paper estimated the impact of political corruption on national poverty through multiple regression and FE models. Model (1) in Table 2 shows the results of an ordinary least squares model (OLS); model (2) shows the results of a feasible generalized most miniature squares model (FGLS), which introduces heteroscedasticity and time effect. Models (3), (4), and (5) are FE models considering different country effects; model (6) incorporates the dependent variables with one lag period and two lag periods into the fixed effects model. All models suggest that the aggravation of political corruption will significantly worsen global poverty, so H1 was confirmed.

In addition, model 6 uses both lagged dependent variables (LDVs) and FE models, and the independent variables still have good significance when the regression coefficients are limited (Angrist & Pischke, 2009). This indicates that there is a time-dependent effect of the model, and previous poverty may affect poverty now or even next year.

The analysis results of other control variables in the FE model are as follows: First, higher tax burden, government expenditure on education, access to electricity, employment in services and the growth of GDP alleviated national poverty. Second, investment freedom, unemployment, population density, and financial freedom worsened national poverty. We also found that the proportion of arable land to agricultural land on national poverty was insignificant in the FE model.

Evidence of moderating and mediating effects from global panel data

The above empirical analysis showed that political corruption exacerbates national poverty, but the empirical models had an endogenous problem generally because of measurement error or interaction variables. This paper took the moderating effect and mediating effect to conduct a heterogeneity analysis to ensure the stability of the model.

The mediating effect. In Table 3, in models (1) and (2), Sobel-Goodman test found that increasing corruption in a country will narrow education equality and reduce the primary education enrollment rate. The decline in primary education enrollment rate was associated with the rise in the incidence of poverty in the country, which meant that political corruption indirectly aggravated national poverty by reducing the primary education enrollment rate. Models

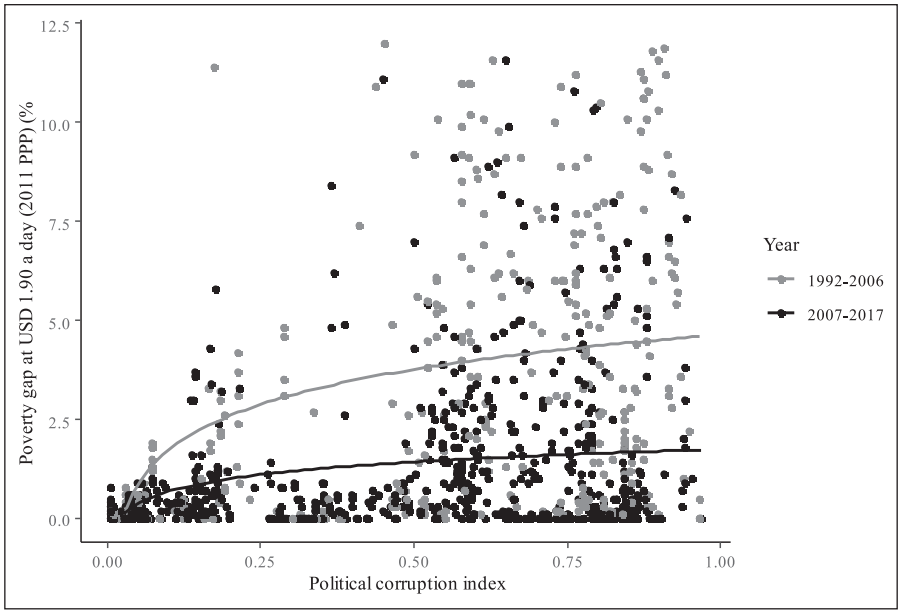

Figure 2. Scatter map of political corruption and national poverty incidence.

Table 2. The baseline regression results.

	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	FGLS	FE	FE	FE	FE
Political corruption index	4.89*** (4.93)	4.52*** (3.84)	7.33*** (3.64)	4.33** (2.13)	5.24*** (2.59)	2.60* (1.83)
Tax burden	−0.09*** (−5.71)	−0.11*** (−5.44)		−0.16*** (−6.83)	−0.14*** (−6.28)	−0.04** (−2.44)
Investment freedom	0.05*** (3.53)	0.02 (1.34)		0.05*** (3.79)	0.00 (0.20)	−0.00 (−0.63)
Unemployment, total (% of total labor force) (national est.)	0.14*** (3.33)	0.08* (1.89)		0.27*** (5.86)	0.20***(4.06)	0.06* (1.78)
Access to electricity (% of population)	−0.34*** (−21.06)	−0.36*** (−22.75)		−0.24*** (−4.72)	−0.16*** (−3.01)	−0.02 (−0.28)
GDP growth (annual %)	−0.02 (−0.47)	−0.04* (−1.73)		−0.07** (−2.36)	−0.07** (−2.14)	−0.08*** (−4.06)
Government expenditure on education, total (% of government expenditure)	0.03 (0.59)	−0.15*** (−2.95)		−0.43*** (−6.86)	−0.38*** (−6.30)	−0.07 (−1.61)
Population density (people per sq. km of land area)	−0.00 (−0.57)	−0.00 (−0.63)		0.03** (2.51)	0.03*** (2.62)	0.01 (0.62)
Financial freedom	−0.00 (−0.26)	0.01 (0.78)		0.03** (2.10)	0.03** (2.37)	0.03** (2.51)
Arable land (% of agricultural land)	−0.04*** (−6.02)	−0.06*** (−5.23)		−0.05 (−0.87)	−0.04 (−0.71)	−0.00 (−0.03)
Population, total	0.00*** (6.84)	0.00*** (7.47)		−0.00*** (−5.83)	−0.00*** (−6.41)	−0.00 (−1.07)
Employment in services (% of total employment) (modeled ILO)	−0.20*** (−9.12)	−0.16*** (−6.15)		−0.44*** (−9.41)	−0.40*** (−6.06)	−0.07 (−1.61)
Constant	50.66*** (20.55)	56.36*** (21.36)	5.51*** (5.51)	67.59*** (12.17)	58.99*** (9.12)	10.57* (1.75)
Observation	617	617	1,319	617	617	445
# of countries	92	92	154	92	92	58
R^2	.72	.76	.01	.60	.66	.89
Country effect	NO	NO	YES	YES	YES	YES
Year effect	NO	NO	NO	NO	YES	YES

Note. *, **, and *** represent the significant confidence of 10%, 5%, and 1% respectively. The *t* statistic of the corresponding explanatory variable is shown in parentheses (same as below).

(3) and (4) show that an increase in the level of corruption in a country increases the Gini coefficient, which was associated with an increase in the incidence of national poverty, implying that political corruption indirectly increases the incidence of poverty in a country by increasing social inequality. As a result, H2a and H2b were confirmed.

Table 3. Mediating effect.

	(1)	(2)	(3)	(4)
	School enrollment, primary (% gross)	Poverty gap at USD 3.20 a day (2011 PPP) (%)	GINI index (World Bank estimate)	Poverty gap at USD 3.20 a day (2011 PPP) (%)
Political corruption index	7.47** (2.33)	5.24*** (2.59)	3.32** (2.12)	5.24*** (2.59)
GINI index (World Bank estimate)				0.48*** (7.56)
School enrollment, primary (% gross)		0.01 (0.22)		
Sobel-Goodman		10.85%		
Mediation tests		−0.01		
Control variables	√	√	√	√
Constant	62.39*** (8.24)	58.99*** (9.12)	31.03*** (6.19)	58.99*** (9.12)
Observation	1,040	599	617	617
# of countries	127	91	92	92
R^2	.11	.66	.43	.66
Country effect	YES	YES	YES	YES
Year effect	YES	YES	YES	YES

Table 4. Moderating effect.

	(1)	(2)	(3)	(4)	(5)	(6)
	Foreign direct investment <6.67	Foreign direct investment >6.67	All data	Public services <5.69	Public services >5.69	All data
Political corruption index	5.18** (2.10)	1.62 (0.45)	5.23*** (2.59)	1.46*** (2.83)	7.75 (1.59)	3.15** (2.56)
Control variables	√	√	√	√	√	√
Constant	54.35*** (7.67)	28.30 (1.09)	59.11*** (9.11)	3.65 (1.01)	37.67** (2.21)	36.63*** (7.12)
Observation	482	135	617	353	127	480
# of countries	85	43	92	60	39	85
R^2	.67	.81	.66	.23	.75	.53
Country effect	YES	YES	YES	YES	YES	YES
Year effect	YES	YES	YES	YES	YES	YES

The moderating effect. Group regression is an effective method to deal with heterogeneity. This paper took the net inflow of foreign direct investment (percentage of GDP) to measure the foreign direct investment and divided all nations into two groups. This takes the median proportion of foreign direct investment in a country (6.67) as a line: countries with low foreign direct investment (<6.67) and countries with high foreign direct investment (>6.67), as indicated by models (1), (2), and (3) in Table 4.

Although political corruption aggravating national poverty was significant in the total sample regression, this effect in countries with a lower foreign direct investment was distinctly more remarkable than in countries with higher foreign direct investment. Countries with a low foreign direct investment need to improve the environment for domestic and foreign investment, take it as a starting point, and alleviate the worsening influence of political corruption on national poverty. Therefore, H3a was confirmed.

We also took the public services as an operational index to measure the level of governance capacity in different countries. According to the average (5.69) of public services in countries, this paper divided all nations into two groups: low public services (<5.69) and high public services (>5.69), which emerged in models (4), (5), and (6) in Table 4. In countries with a low level of public services, the effect of political corruption aggravating national poverty was greater than that of countries with high levels of public services. The results suggested that improving governance capacity could effectively alleviate the worsening influence of political corruption on national poverty. Therefore, H3b was confirmed.

Substitution variable method and endogenous test

Considering the diverse characteristics among 117 countries in terms of politics, economy, culture, and society, which may lead to the instability of the model results, to ensure the reliability of the above research results, we adopted variable substitution and instrumental variables to test robustness.

Table 5. Robustness test based on FE model.

	(1)	(2)	(3)	(4)
	FE	FE	FE	FE-IV
Political corruption index	3.84*** (2.79)	2.06* (1.90)	2.04** (1.98)	6.60** (2.19)
Control variables	√	√	√	√
Constant	2.35*** (3.43)	35.91*** (12.09)	25.94*** (7.86)	60.83*** (7.91)
Observation	1,319	617	617	548
# of countries	154	92	92	79
R^2	.01	.54	.64	.54
Country effect	YES	YES	YES	YES
Year effect	NO	NO	YES	YES
Sargan test				1.39 (0. 24)
Davidson–MacKinnon test				0.10 (0. 75)

First, this paper took the poverty gap at USD 1.90 a day as the dependent variable for regression analysis. The FE models in Tables 2 and 5 are consistent in direction and significance, indicating that those conclusions mentioned previously are robust.

Secondly, the authors chose patent grants and political corruption (lag one step) as instrumental variables for verification. In model (4) of Table 5, the insignificance of the Sargan test results in this paper indicates that instrumental variables were strictly exogenous. Meanwhile, the test results implied significant endogenous problems, so applying the instrumental variable method based on FE was necessary. At the same time, the regression results of IV-FE still hinted that political corruption had a significant positive impact on the incidence of national poverty, which was consistent with previous conclusions and further verifies their robustness.

Conclusions

Based on the global panel data of 117 countries from 1992 to 2017, this paper used multiple regression and FE models to explore how political corruption affected worldwide poverty and further verified the research results through a robustness test. At the same time, different regression models and instrumental variables' models were applied to increase the reliability and validity of the empirical results. The empirical analysis showed that: (1) There was a significant positive correlation between political corruption and national poverty; (2) Primary education and the Gini index played a mediating role between political corruption and national poverty; (3) The level of foreign direct investment and public services moderated the relationship between political corruption and national poverty.

The novelty of this study is reflected in the theory and method. We took the mediating effect to confirm the educational model and polarization model of political corruption worsening national poverty, and the moderating effect to confirm the economic and governance model. Meanwhile, in terms of research methods, this paper combined multiple regression and FE models for empirical analysis and chose the instrumental variable method for endogenous testing, which increased the robustness of the empirical model.

Meanwhile, the findings could provide policy implications for reducing national poverty. Firstly, national poverty occurs in less developed countries, and developing countries can attract more foreign direct investment by improving the business environment to promote local economic development. They can improve government efficiency and management ability with good public services. This is an excellent way to reduce national poverty through economic and governance reforms. Second, providing good primary education and reducing social inequality mitigates the harmful effects of political corruption and the incidence of national poverty. Thus, effective national poverty reduction strategies and policies must look beyond poverty to the more indirect causes of national poverty.

Of course, this paper also has research limitations. For instance, it is impossible to incorporate more control variables to reflect differences in political systems and economic development levels among countries because of the data restrictions. Follow-up research needs to improve and supplement this aspect to deepen the understanding of research in this field.

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Supplemental material

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